

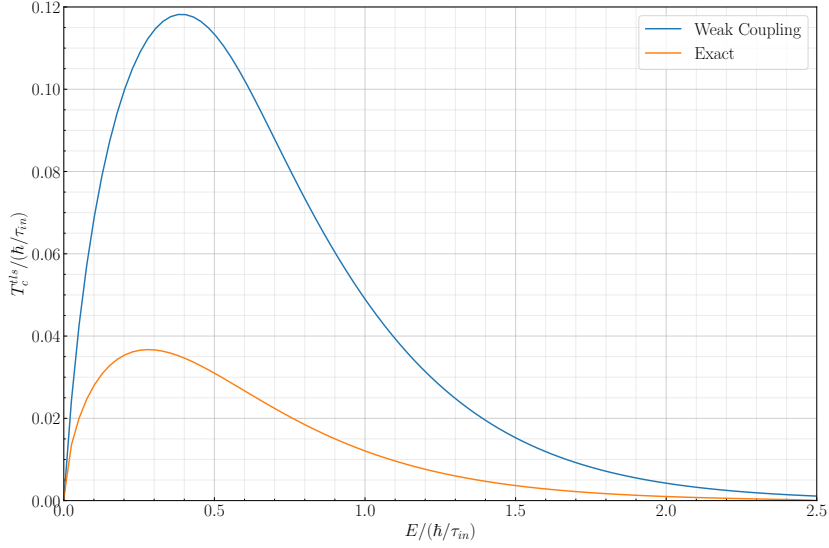


FIG. 7. Superconducting transition temperature, T_c , induced by electron-TLS scattering versus tunnel splitting, E . Both T_c and the tunnel splitting are scaled in units of the energy associated with the inelastic scattering rate, $\hbar/\tau_{in}$. The “Weak Coupling” result based on Eq. (72) substantially overestimates T_c for electron-TLS mediated superconductivity.

where $\tilde{\varepsilon}_n = \varepsilon_n + i\Sigma_\epsilon(\varepsilon_n)$, with

$$i\Sigma_\epsilon(\varepsilon_n) = \frac{1}{2\pi\tau_{in}} \int dE p(E) N_{ge}(E) \pi T \sum_{\varepsilon_{n'}} \mathfrak{D}(\varepsilon_n - \varepsilon_{n'}; E) \frac{\varepsilon_{n'} + i\Sigma_\epsilon(\varepsilon_{n'})}{\sqrt{(\varepsilon_{n'} + i\Sigma_\epsilon(\varepsilon_{n'}))^2 + \tilde{\Delta}(\varepsilon_{n'})^2}}. \quad (64)$$

The onset of superconductivity for electron-TLS mediated pairing is obtained by linearizing Eq. (63) in $\tilde{\Delta}(\varepsilon_n)$, and setting $\tilde{\Delta}(\varepsilon_n) = 0$ in Eq. (64) for Σ_ϵ . The resulting homogeneous equation for $\tilde{\Delta}$ is

$$\tilde{\Delta}(\varepsilon_n) = \frac{1}{2\pi\tau_{in}} \int dE p(E) N_{ge}(E) \pi T \sum_{\varepsilon_{n'}} \mathfrak{D}(\varepsilon_n - \varepsilon_{n'}; E) \frac{\tilde{\Delta}(\varepsilon_{n'})}{|\varepsilon_{n'} + i\Sigma_\epsilon(\varepsilon_{n'})|}, \quad (65)$$

where Σ_ϵ reduces to

$$i\Sigma_\epsilon(\varepsilon_n) = \frac{1}{2\pi\tau_{in}} \int dE p(E) N_{ge}(E) \pi T \sum_{\varepsilon_{n'}} \mathfrak{D}(\varepsilon_n - \varepsilon_{n'}; E) \text{sgn}(\varepsilon_{n'}). \quad (66)$$

Equation (65) can be expressed as a matrix eigenvalue equation in the space of Matsubara frequencies with indices (n, n') ,

$$\sum_{n'} (K_{n,n'} - \delta_{n,n'}) \tilde{\Delta}(\varepsilon_{n'}) = 0, \quad (67)$$

where

$$K_{n,n'} = \frac{\hbar}{2\pi\tau_{in}} \int dE' p(E') N_{ge}(E') \pi T \mathfrak{D}(\varepsilon_n - \varepsilon_{n'}; E') \frac{1}{|\varepsilon_{n'} + i\Sigma_\epsilon(\varepsilon_{n'})|} \Big|_{T_c^{tls}}. \quad (68)$$

is the kernel of the pairing self energy. If a non-trivial solution exists with $|\tilde{\Delta}(\varepsilon_n)| \xrightarrow{T \rightarrow T_c^{tls} - 0^+} 0^+$ for $T_c^{tls} > 0$, then the superconducting transition temperature is determined by the vanishing determinant,

$$\text{Det}(K_{n,n'} - \delta_{n,n'}) \Big|_{T_c^{tls}} = 0. \quad (69)$$

For TLS impurities with a delta function distribution, $p(E') = \delta(E' - E)$, i.e. all TLS impurities with the same tunnel splitting E , Eq. (68) simplifies to

$$K_{n,n'} = \frac{\hbar}{\tau_{in}} N_{ge}(E) T_c^{tls} \frac{E}{E^2 + (\varepsilon_n - \varepsilon_{n'})^2} \frac{1}{|\varepsilon_{n'} + i\Sigma_\epsilon(\varepsilon_{n'})|} \Big|_{T_c^{tls}}. \quad (70)$$

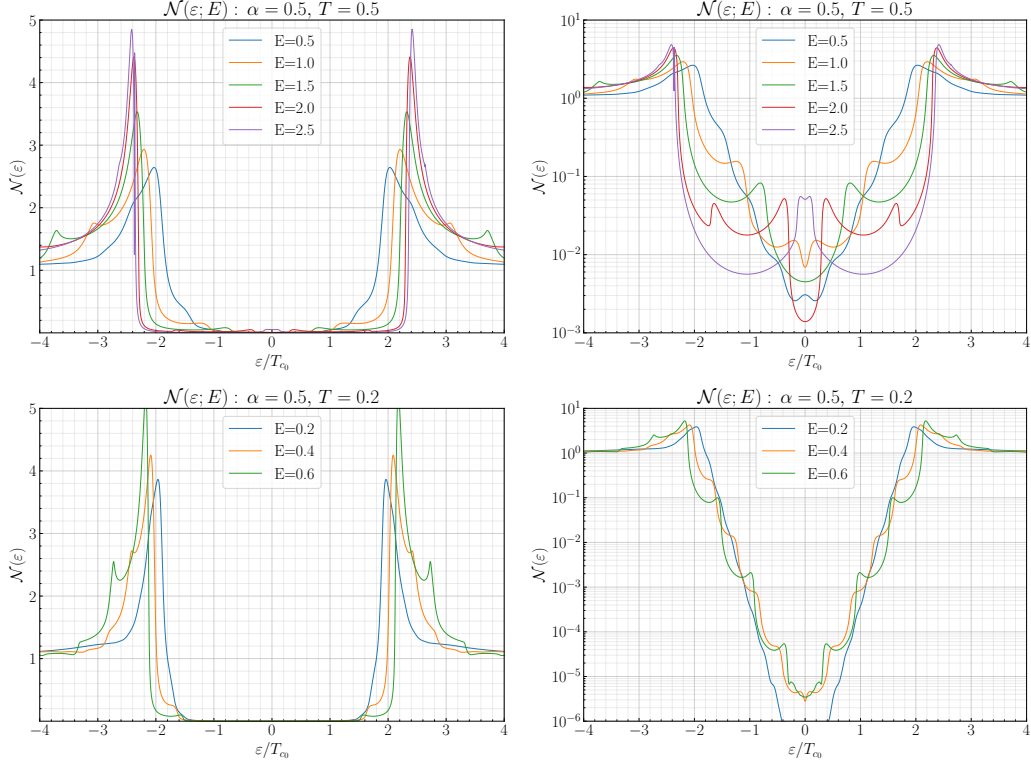


FIG. 8. The quasiparticle DOS for several delta function distributions. The top row is the equilibrium DOS for $T = 0.5 T_{c0}$, linear and log scales, showing sub-gap quasiparticle states and above gap resonances. The bottom row corresponds to $T = 0.2 T_{c0}$, and illustrates the sensitivity of the subgap DOS to level population of the TLS impurities.

The kernel decays rapidly for $|\varepsilon_n|, |\varepsilon_{n'}| \gg E$. Thus, we introduce a cutoff $|\varepsilon_{n,n'}| \leq E_c$ with $E_c \geq E$, then evaluate the determinant in Eq. (69) as a function of $\hbar/\tau_{in}$ and E , and locate the zeroes.

In the limit $E \gg 2\pi T_c^{tls}$ we replace the Bosonic propagator by $E/(E^2 + (\varepsilon_n - \varepsilon_{n'})^2) \rightarrow 1/E$ for $|\varepsilon_n|, |\varepsilon_{n'}| \leq E$, which is analogous to the weak-coupling limit for phonon-mediated superconductivity, but with E analogous to the maximum phonon frequency. In this limit $\tilde{\Delta}(\varepsilon_n) = \Delta\Theta(E - |\varepsilon_n|)$, in which case $i\Sigma_\varepsilon = 0$, and thus the linearized gap equation becomes,

$$1 = \frac{\hbar}{\tau_{in} E} N_{ge}(E) T_c^{tls} \sum_n^{| \varepsilon_n | \leq E} \frac{1}{|\varepsilon_n|} \Big|_{T_c^{tls}}. \quad (71)$$

Evaluating the diGamma function (see paragraph after Eq. (47)) in the limit $E \gg T_c^{tls}$ gives the transcendental equation for T_c^{tls} ,

$$T_c^{tls} = \frac{2e^\gamma}{\pi} E \exp\left\{-\frac{\pi E}{N_{ge}(E) \hbar/\tau_{in}}\right\} \approx 1.13 E \exp\left\{-\frac{\pi E}{N_{ge}(E) \hbar/\tau_{in}}\right\}. \quad (72)$$

Note that the difference in populations of the ground and excited states enters the exponent, and as a result the solution for T_c^{tls} is very sensitive to the relative level populations. For $N_{ge} > 0$ there is always a superconducting transition. However, there is no physical solution for $N_{ge} \leq 0$. The weak-coupling formula, Eq. (72), for T_c^{tls} is plotted in Fig. 7. Note that at the peak value of T_c^{tls} we have $E/2\pi T_c^{tls} \approx 0.5$, in which case the assumption underlying weak-coupling approximation is far from being satisfied. This is born out by numerical solutions of Eq. (69) with the kernel in Eq. (70) with the results also shown in Fig. 7. The numerical determination of T_c^{tls} is obtained by choosing a cutoff energy $E_c \gg \max\{2\pi T, E\}$ that defines the maximum Matsubara energy, or $N_c = \text{Int}(E_c/2\pi T)$, i.e. the dimension of matrix $K_{n,n'}$. We find that for $N_c = 10^3$ the numerical result for T_c^{tls} is converged and insensitive to extending the dimension of the matrix $K_{n,n'}$.

It is interesting to consider the possibility that TLS disorder introduced into clean non-superconducting metals such as Au, Cu, Pt etc. might lead to superconductivity, but it is also important to note that repulsion from the Coulomb pseudo potential may compensate the attractive interaction mediated by TLS impurities.

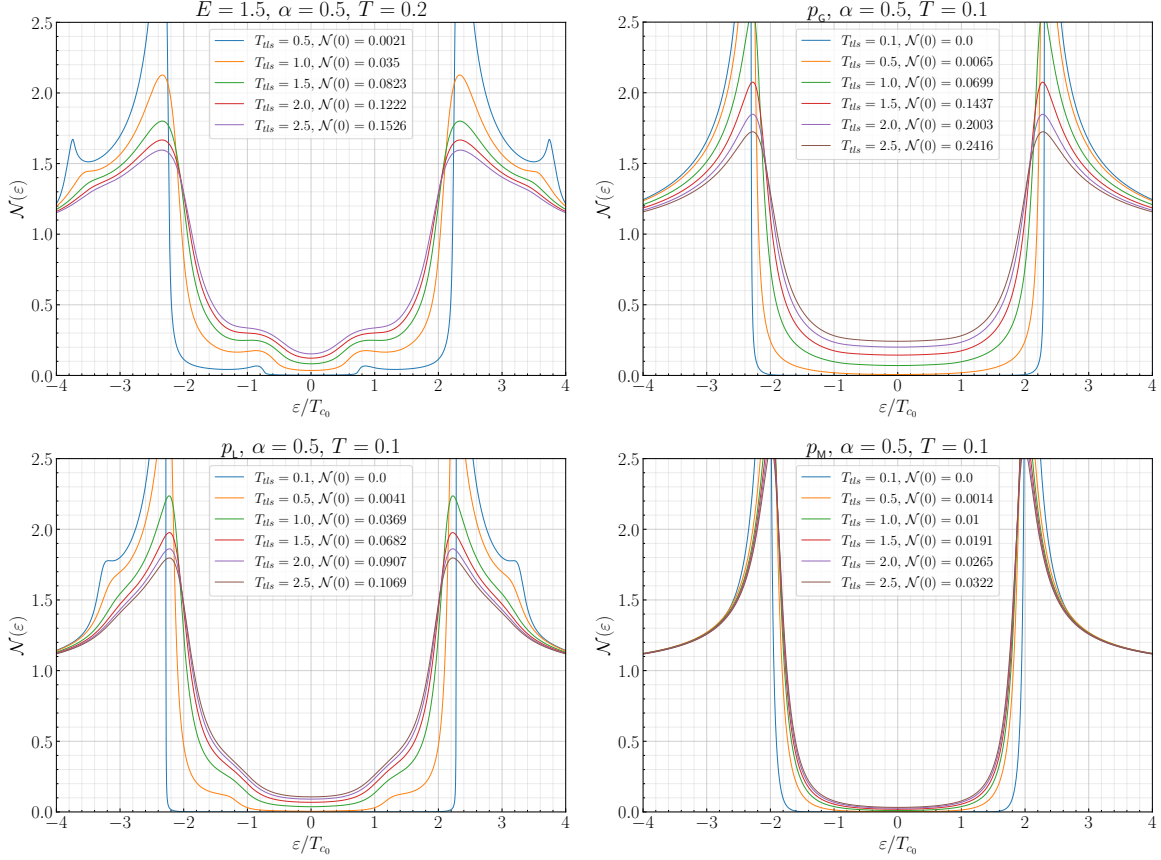


FIG. 9. The quasiparticle DOS for four TLS distributions: delta function (top left), glass-like (top right), Lorentzian (bottom left) and modified Wipf-Neumaier (bottom right). The level populations of the TLS impurities are out of equilibrium with the thermal bath, and described by the effective temperature T_{tls} . Pair-breaking by the non-equilibrium TLS population leads to a gapless spectrum for all TLS models.

VI. SUB-GAP STATES, PAIR-BREAKING & NONEQUILIBRIUM TLS LEVEL POPULATIONS

Solving the real-frequency gap equation and retarded self energies is computationally expensive compared to the Matsubara formulation. However, it has the advantage that the spectrum of electronic states contributing to the order parameter and the self energy are obtained explicitly, and thus separate from the distributions of occupied and unoccupied quasiparticle energies and TLS levels.

A. Density of states

In particular, the quasiparticle density of states, $\mathcal{N}(\varepsilon)$, for $T < T_c$ is computed from Eqs. (53), and (56), once we have calculated the order parameter and the retarded self energies (55). Inelastic scattering of quasiparticles that are bound to form Cooper pairs leads both pair breaking in addition to enhancement of superconductivity. For the TLS impurities with level populations in equilibrium with the thermal bath, pair breaking is weaker than pair enhancement. Nevertheless, subgap quasiparticle states indicative of pair breaking appear in the quasiparticle density of states as shown in Fig. 8. The first row corresponding to $T = 0.5T_{c0}$ shows subgap states at all energies down to the Fermi level ($\varepsilon = 0$), with $\mathcal{N}(0)/N_f \sim 0.01 - 0.1$, as well as above-gap resonances, particularly for the largest tunnel splitting, $E = 2.5T_{c0}$. The spectral weight of the sub-gap states is suppressed at lower temperatures, e.g. $T = 0.2T_{c0}$ shown in the second row, as the TLS impurities are more likely to be in the ground energy level.

Enhanced Pair-Breaking by Nonequilibrium TLS Level Populations

The sensitivity of the sub-gap spectrum of quasiparticles and pairing self energy to the relative populations of the excited and ground states of the distribution of TLS impurities also implies that ineffective equilibration of the TLS population to the conduction electrons bath and the lattice can have substantial impact on the